

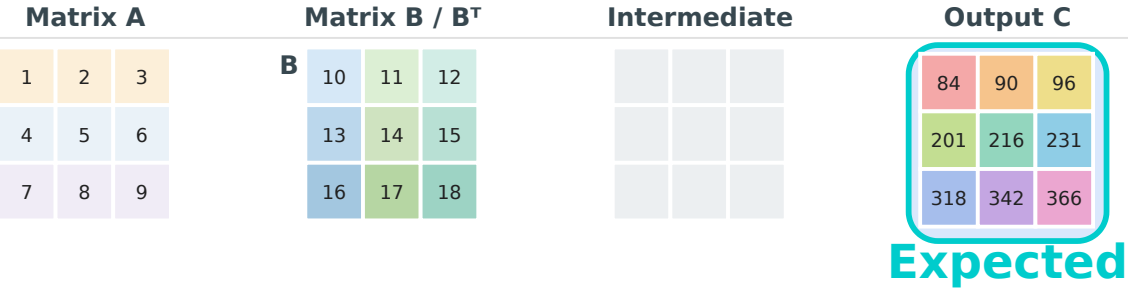
matmul_8bitraw_short — C = A × B on the T1 fabric (3×3 worked example)

Steps run top → bottom. Colour key: A by row; B by column-hue (so the V-load transpose turns columns into B^T rows); each Output cell keeps its final-position colour, so a result enters at column 0 and shifts right into place.

Example

0. Reference

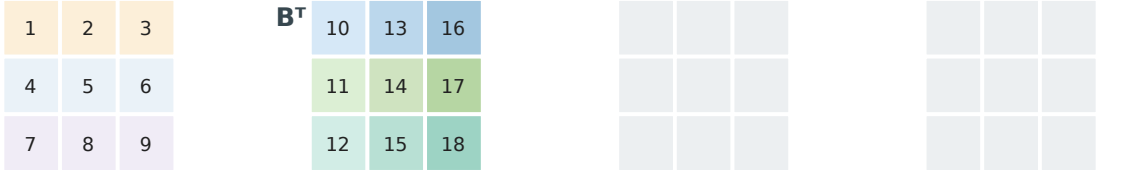
C = A × B (true result)



Setup

1. Load H / V

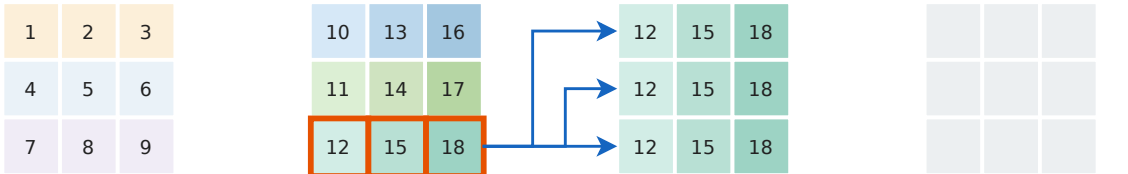
vle8 v4,(A) [H]
csrwi 0x7c0,1
vle8 v5,(B) [V] → B^T



Iteration 0 (Full Steps)

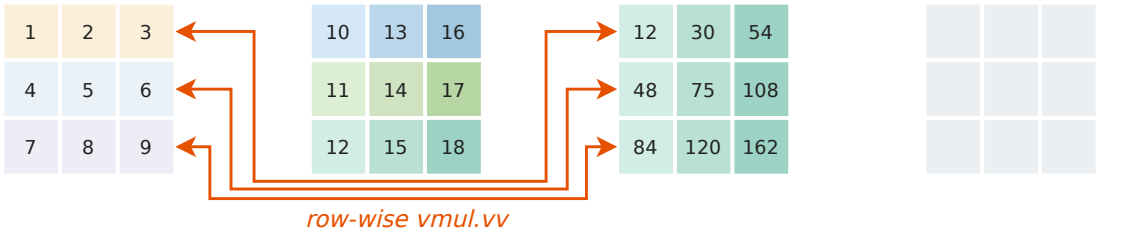
2. k=2 Broadcast

csrwi 0x7c0,1
vrgather.vx v6,v5,2 [V]



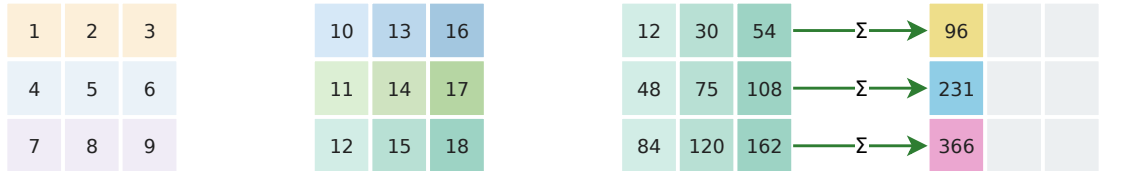
3. k=2 Multiply

csrwi 0x7c0,0
vmul.vv v6,v4,v6 [H]



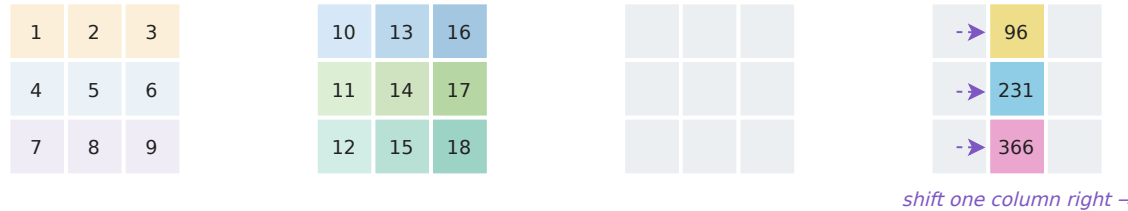
4. k=2 Reduce

vredsum.vs v7,v6,v28 →
v7[0]=C*[2]



5. Shift →

vslideup.vi v8,v7,1
vmv.v.v v7,v8

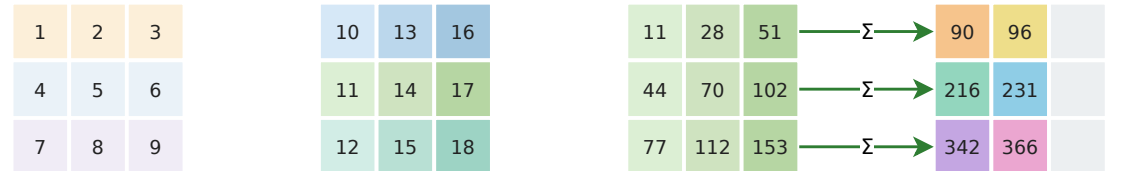


SKIP 6. Broadcast, 7. k=1 Multitply

Iteration 1 (Simplified)

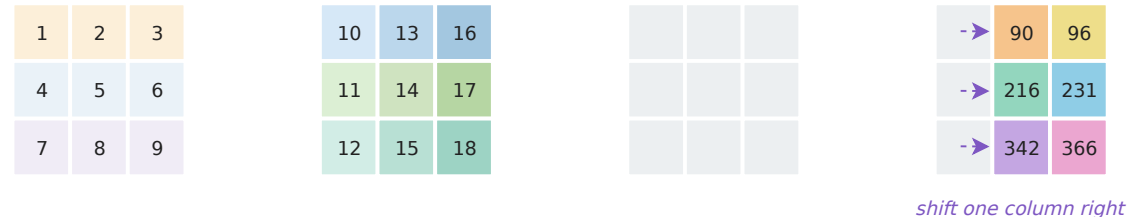
8. k=1 Reduce

vredsum.vs v7,v6,v28 →
v7[0]=C*[1]



9. Shift →

vslideup.vi v8,v7,1
vmv.v.v v7,v8



SKIP 10. Broadcast, 11. k=0 Multitply

Iteration 2 (Sim)

12. k=0 Reduce

vredsum.vs v7,v6,v28 →
full C
vse8 v7,(C)

